Saqib Bedar

Software Engineer | Python, Backend & Full-Stack Development

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PROFESSIONAL SUMMARY

Software Engineer focused on Python, backend architecture, and full-stack development with an applied ML research trajectory. Experienced in RESTful API design, authentication architectures (4-role RBAC, JWT rotation, session security), and developer tools using modern engineering practices (uv, Ruff, pytest). Conducting final-year undergraduate research on latent-predictive vision-language representation learning (JEPA) for low-resource languages under faculty supervision at Quaid-i-Azam University.

TECHNICAL SKILLS

Python & Backend:	Python, FastAPI, Pydantic, REST API Design, OOP, Async Programming, uv, Ruff, pytest
Databases & Cloud:	MongoDB, MySQL, PostgreSQL (learning), GCP (learning), Data Modeling
Full-Stack Web:	JavaScript, React, Node.js, Express, Tailwind CSS, HTML & CSS, REST APIs, Authentication & JWT
ML & Applied AI:	PyTorch, Vision-Language, Computer Vision, Transfer Learning, Model Evaluation, NumPy, Pandas, OpenAI & Gemini SDKs
Systems & Tooling:	Git, GitHub, GitLab, GitHub Actions, Linux/Ubuntu, Docker, Technical Documentation, C++

EXPERIENCE & TEACHING

Teaching Assistant - Web Application Development *Quaid-i-Azam University, Islamabad* **2026 - Present**

Support undergraduate computer science students under the supervision of Dr. Rabeeh Abbasi. Conduct hands-on lab sessions teaching modern web application development, Git repository workflows, RESTful APIs, and SDK integrations. Assist with code reviews, student debugging support, and comprehensive academic evaluation reports.

Front-End Developer Intern *College of Army Education (CAE)* **2025**

Completed an 8-week front-end development internship, designing and implementing responsive user interfaces using HTML, CSS, JavaScript, and modern UI practices within a structured software engineering environment.

Speaker - Open Source Contribution Webinar *AutoMind Community* **Sep 2026**

Delivered an interactive technical webinar via Google Meet on open-source software contribution, covering Git fundamentals, branch management, collaborative workflows, and pull request hygiene.

Speaker & Workshop Instructor *MLSA QAU* **Sep 2024**

Conducted the 'GitHub: Learn What Matters' technical workshop via Microsoft Teams, guiding university students through version control essentials, branching strategies, conflict resolution, and GitHub Pages deployment.

SELECTED PROJECTS

UV-JEPA *Final-Year Research FYP (Supervised by Dr. Syed Muhammad Naqi, QAU)* **2026 - Present**

Investigating latent-predictive vision-language representation learning (JEPA) for low-resource languages, evaluated on Urdu under data-constrained settings. Architected dual encoders (ViT-Small and RoBERTa) with an EMA target text encoder and cross-attention predictor trained via normalized latent regression to achieve data efficiency without autoregressive decoding errors. Benchmarked against tuned contrastive and autoregressive baselines.

Stack: PyTorch, Vision Transformer (ViT), RoBERTa, Cross-Attention, JEPA, Representation Learning

Online Student Portal System *Production Client Work* **Dec 2025**

Engineered an enterprise university management portal featuring hierarchical 4-role access control (Master Admin, Department Admin, Faculty, Student). Built secure JWT authentication with HTTP-only cookies, automated refresh-token rotation, server-managed concurrent session limitation, and rate-limited OTP password recovery across a modular MVC backend.

Stack: Node.js, Express, React, MongoDB, JWT Auth, RBAC, Tailwind CSS

Fluent Python *Open-Source Python Learning Hub* **2025 - Present**

A structured open-source Python engineering resource covering language internals, object-oriented design, context managers, async programming, and LLM integrations. Engineered with modern tooling standards using uv for environment management, Ruff for linting, and automated pytest test suites.

Stack: Python 3.12+, uv, Ruff, pytest, AsyncIO, OOP, Concurrency

Date Fruit Classification *Applied Machine Learning* **2024**

Comparative computer vision study evaluating ResNet-50 transfer learning against a custom CNN trained from scratch on 7,000 images across 10 classes. Achieved 93.96% test accuracy and deployed a lightweight PyTorch web inference application for real-time model evaluation.

Stack: 93.96% Accuracy, ResNet-50, Custom CNN, PyTorch, Transfer Learning, Computer Vision

Gitcraft *Technical Documentation Site* **2025**

Authored a comprehensive 33-page technical Git documentation site built with Zensical. Details practical Git mental models, branching strategies, collaborative workflows, and disaster recovery patterns.

Stack: 33 Pages, Zensical, Git, Technical Documentation, GitHub Pages

EDUCATION

BS in Computer Science *Quaid-i-Azam University, Islamabad* **2023 - Present**

Core Coursework: Data Structures, Analysis of Algorithms, Operating Systems, Database Systems, Computer Networks, Artificial Intelligence, Computer Vision, Theory of Automata.

COURSES & VERIFIED CERTIFICATES

Introduction to Networking | *NVIDIA* May 2025
C++ Basics: Selection and Iteration | *Codio* March 2025
Tools of the Trade: Linux and SQL | *Google* May 2025
Google AI Essentials | *Google* August 2024
Programming with JavaScript | *Meta* August 2023